AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

CSE 4203 — Computer Graphics

TABLE OF CONTENTS / QUESTION-PAPER INDEX

Question papers: 3 Main numbered questions: 9 Source pages: 10

Examination	Main questions	Question numbering	PDF / DOCX pages
Spring 2025	3	1–3	2–5
Fall 2024	3	1–3	6–8
Spring 2024	3	1–3	9–11

How the questions are counted

A main question means a printed “Question 1”, “Question 2”, etc. Lettered and Roman-numbered subparts are not counted separately. Where numbering restarts in Part B, those questions are counted separately. The paper count follows the semester labels in the original source.

Edition and source notes

• Question text and numeric tables are editable; original diagrams and complex notation are retained as image excerpts.

• Headers, black-and-white styling, right-margin marks and source-page breaks have been restored. Exact font metrics and line breaks can vary in Word.

• Papers are ordered newest to oldest by the printed semester. Fall 2024 (three pages) was added from CG-semFinal.pdf. Spring 2025 and Spring 2024 were already present and are included once only.

• Fall 2024: printed wording, dimensions and the table heading “RBGA” are retained as supplied. Handwritten working is not part of the typed text; annotations within the original half-circle figure remain.

The index page is additional. On the question pages, the footer retains the source paper’s page reference. The ranges above refer to this complete PDF/DOCX, including this index.

Ahsanullah University of Science and Technology
Department: Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering
Semester Final Examination: Spring 2025
Year: 4th Semester: 2nd
Course Number: CSE4203
Course Title: Computer Graphics

Time: 3 (Three) hours **Full Marks: 70**

Instruction: There are three sets of questions. Answer any five questions from each set. Marks allotted are indicated in the margin.

Question 1: Answer any 5 questions. **[4×5=20]**

a) Identify the problems associated with the lighting model defined by the equation below and propose the standard mathematical correction to resolve it. [4]

$$c = c_r c_l \, | n \cdot l |$$

b) Suppose engineers are programming a basic graphical interface for a low-power embedded medical device that consists of a basic microcontroller and an LCD screen. Justify which graphics pipeline is most suitable for this device. [4]

c) Describe the fundamental topological advantage of using triangles for surface modeling over higher-order polygons. [4]

d) State the differences between image-order and object-order rendering. [4]

e) Explain different kinds of memory and their roles inside a GPU. [4]

f) Explain which of raster or vector graphics is better for images with minimal color variation, such as flat interface icons. [4]

g) Describe why perspective projection is considered a non-affine transformation. [4]

Question 2: Answer any 5 questions. **[5×5=25]**

a) Show that two successive reflections about either of the principal axes is equivalent to a single rotation about the coordinate origin. [5]

b) Apply appropriate changes to express the perspective matrix implemented in OpenGL in terms of field of view, aspect ratio, near plane, and far plane. [5]

c) A uniform quadratic B-Spline curve S is defined by 6 control points $P_0(-5, 0)$, $P_1(-2, 5)$, $P_2(0, -1)$, $P_3(3, 6)$, $P_4(5, -2)$ and $P_5(10, -5)$. Find the point on the 2nd and 4th curve segments for $t = 4/5$. [5]

d) Consider 3 images img1, img2 and img3 overlapping each other, where img1 is the foreground of img2 and img2 is the foreground of img3. Additionally, img1 has an alpha mask α_1 and img2 is fully opaque. Find the value of α_1 if we want to get the output image as img4. [5]

192	150
200	180
100	120

img1

80	50
100	90
50	60

img2

55	102
64	120
75	105

img3

180	100
135	155
90	90

img4

e) Determine whether Polygon A and Polygon B should be rendered or discarded during the backface culling stage of the graphics pipeline, given the viewer's eye position is located at (4, 3, 8). Parameters of the polygons are given below: [5]

Polygon A: Contains a surface point $P_a(2, 1, 0)$ and surface normal vector $N_a(-1, -1, 2)$.

Polygon B: Contains a surface point $P_\beta(2, 5, -3)$ and surface normal vector $N_\beta(1, 2, -1)$.

f) Consider an image with the pixel values shown below. Each pixel can hold up to 24 bits of data. Apply run-length encoding (RLE) in two different ways: by scanning the image row-wise (horizontally) and column-wise (vertically). For each method, calculate the compression ratio and determine which method gives better compression. Assume 1 byte is allocated to store the frequency count for each run. [5]

12	12	12	45	45	78	78
12	12	12	45	45	78	78
90	90	33	33	33	33	33
90	90	33	33	33	33	33
10	10	10	10	50	50	50
10	10	10	10	50	50	50

g) Consider a hexagonal object with vertices A(3, 3), B(5, 3), C(6, 5), D(5, 7), E(3, 7) and F(2, 5). The object is rotated by 50 degrees clockwise about its centroid, followed by 130% scaling. Find the composite transformation matrix to do the task. [5]

Question 3: Answer any 5 questions. **[5×5=25]**

a) Consider a triangle with vertices A(20, 10), B(25, 15), and C(22, 20), and normals $N_a(0, 0, 1)$, $N_\beta(0.8, 0, 0.6)$ and $N_c(0, 0.8, 0.6)$ at each vertex of the triangle. To achieve smooth shading and eliminate faceted appearance, the vertex normals need to be interpolated across the polygon to generate per-fragment normals, using the exact same method used for color interpolation. Apply barycentric interpolation to find the interpolated normal at P(24, 16) inside the triangle. [5]

b) Consider a rectangle defined by O(0, 0), A(8, 0), B(8, 10) and C(0, 10). Reflect P(3, −3) along the diagonal OB and find the reflected point after applying transformations. [5]

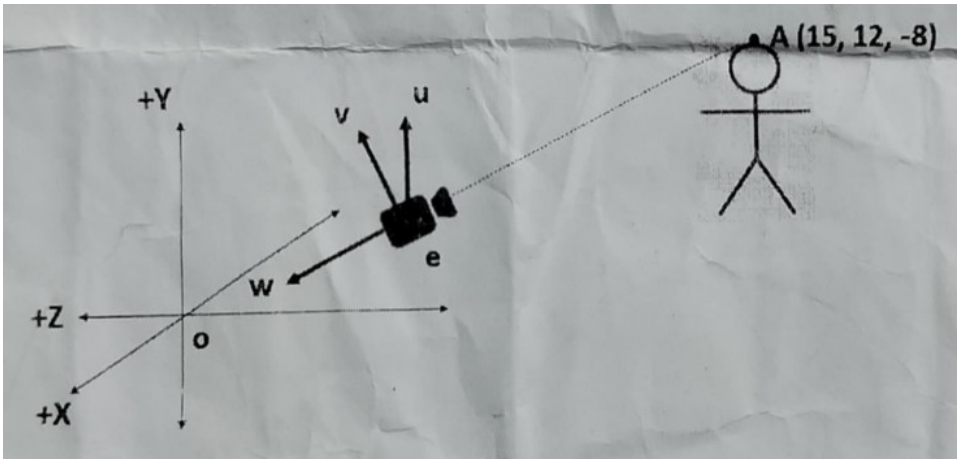
c) A 2 × 4 grayscale image IMG was compressed using the Haar Discrete Wavelet Transform. The compression process involves a row-wise transform, followed by a column-wise transform and a thresholding step. The resulting compressed coefficient matrix is provided below. [5]

39	48	0	0
0	6	0	0

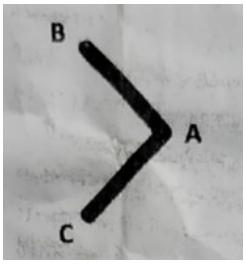
Reconstruct the original image by applying the Inverse Haar Discrete Wavelet Transform (IDWT) to matrix C. Show all intermediate calculations and the matrices obtained at each stage of the reconstruction process.

d) Consider a 3D scene where a line segment AB is defined in a canonical view volume. Instead of the standard canonical view volume, the line is defined in a normalized space defined by (x, y, z) ∈ [−3, 3]. This normalized space is then mapped to a viewport of size 512 × 256, where the origin is located at the top-left corner, with the x-axis increasing to the right and the y-axis increasing downward. Apply the transformation to determine the final screen coordinates of A(2.2, 0.5, 1) and B(1.3, −2.5, −2). [5]

e) Here (in the figure of a shopping mall), origin O and basis {x, y, z} construct a 3D canonical coordinate system. A 3D surveillance camera is installed inside the mall at point e(2, 5, −1) with a basis {u, v, w}. The camera is oriented to face point A to track people entering the mall. Determine the basis vectors of the surveillance camera. Assume the world's up direction is defined by the vector (0, 1, 0). [5]



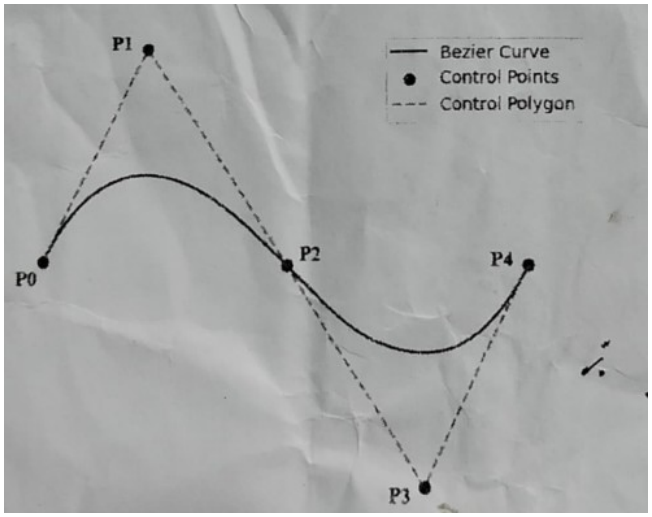
f) Consider a right-pointing chevron symbol (>) created for a user interface, consisting of two line segments joined at a vertex (see figure). The upper line segment AB is rasterized using the Midpoint Line Drawing Algorithm, starting from A(14, 11) and ending at B(11, 16). The lower line segment AC is obtained by reflecting the calculated pixels along the line y = 11. Apply the appropriate method to find the intermediate pixel coordinates for rasterizing this symbol. [5]



Necessary adjustments of the original line drawing algorithm for different octants are provided below:

(1) plot(x, y)	(2) swap(x, y); plot(y, x)	(3) x=−x; swap(x, y); plot(−y, x)	(4) x=−x; plot(−x, y)
(5) x=−x; y=−y; plot(−x, −y)	(6) x=−x; y=−y; swap(x, y); plot(−y, −x)	(7) y=−y; swap(x, y); plot(y, −x)	(8) y=−y; plot(x, −y)

g) A 2D Bezier curve Q is defined by the control points P_0, P_1, P_2, P_3, P_4 (see figure), where the curve is symmetric about a line x = 1 and control point P_2 is used twice. If P_0, P_1 and P_2 are (−6, 1), (−3, 7) and (1, 1) respectively, compute the value of Q(1/5). [5]



Date of Examination: 5/11/2025

Ahsanullah University of Science and Technology
Department: Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering
Semester Final Examination: Fall 2024
Year: 4th Semester: 2nd
Course Number: CSE4203
Course Title: Computer Graphics

Time: 3 (Three) hours **Full Marks: 70**

Instruction: There are three sets of questions. Answer any five questions from each set. Marks allotted are indicated in the margin.

Question 1: Answer any 5 questions. **[3×5=15]**

a) Describe why triangles are commonly used as the primary primitive in computer graphics. [3]

b) Explain the role of the reflection vector (r) in calculating specular highlights. Show how (r) is calculated using the light vector (l) and the surface normal (n). [3]

c) Briefly describe different kinds of hardware graphics pipelines. [3]

d) Explain the two distinct specialized GPU cores optimized for rendering complex 3D scene that involves millions of triangles and lighting effects. [3]

e) Discuss the primary challenges associated with rotation matrices and Euler angles. State the possible solution to overcome these issues. [3]

f) Explain how to determine whether a polygon is facing towards or away from the camera. [3]

g) State the differences between Bezier and B-Spline curves. [3]

Question 2: Answer any 5 questions. **[5×5=25]**

a) Assume M is a composite transformation matrix of a 45 degree rotation about the y-axis followed by a translation of (3, −2). Determine the inverse composite transformation matrix M^{-1}. [5]

b) Using the Phong Shading Model, calculate the highlight intensity for a point on the surface given the light vector l = (1, 1, 0), surface normal n = (0, 1, 0) and eye vector e = (0, 1, 1). Assume that Phong exponent p = 2 and light intensity C_l = (1, 1, 1). [5]

c) Show that, in case of the midpoint line drawing algorithm, we can successively update the decision variable by adding $(y_1 - x_1) - (y_0 - x_0)$ for each selection of a northeast pixel. Here, (x_0, y_0) and (x_1, y_1) are two endpoints of the line. [5]

d) Consider a perspective viewing volume defined by a near plane at n = 2, a far plane at f = 10, a Field-of-View (FoV) of 90° and an aspect ratio of 2:1. Calculate the symmetric frustum boundaries (l, r, b, t) and construct the complete perspective projection matrix. [5]

e) Show the geometry up to iteration 3 and find the fractal dimension generated by the following rules: [5]

Step 1: Start with a square.

Step 2: Subdivide it into 9 equal smaller squares.

Step 3: Keep only the center square and the 4 squares in the middle of each edge. Remove rest of the squares.

Step 4: Repeat this process for each remaining square at the next iteration.

f) Consider a clipping rectangle with its lower-left corner at (3, 4), a width of 12 and a height of 6. For a line segment defined by the points (1, 2) and (18, 9), use the Cyrus-Beck algorithm to find its intersection point with the right edge of the clipping rectangle. [5]

g) Consider a uniform cubic B-Spline curve S defined by control points $P_0(-3, 0)$, $P_1(-1, 2)$, $P_2(0, 4)$, $P_3(2, 3)$, $P_4(3, 1)$, $P_5(4, -1)$ and $P_6(5, 0)$. Construct the matrix format for this curve from the blending function given below and find the points on the curve segments for $S_1(0.2)$ and $S_2(0.8)$. [5]

$B_0(t) = \frac{1}{6}(-t^3 + 3t^2 - 3t + 1)$
$B_1(t) = \frac{1}{6}(3t^3 - 6t^2 + 4)$
$B_2(t) = \frac{1}{6}(-3t^3 + 3t^2 + 3t + 1)$
$B_3(t) = \frac{1}{6}(t^3)$

Question 3: Answer any 5 questions. **[6×5=30]**

a) Render a 4 × 4 pixel scene with three overlapping rectangles. Determine visibility with Z-buffer algorithm and compute the frame buffer. Find the polygon data in the following table: [6]

Polygon	Depth (z)	Color (RBGA)	Pixel Coverage
A	0.5	(1, 0, 0, 0.6)	(1, 1), (1, 2), (2, 1), (2, 2)
B	0.3	(0, 1, 0, 0.7)	(1, 1), (1, 2), (2, 1), (2, 2)
C	0.4	(0, 0, 1, 0.5)	(2, 1), (2, 2), (3, 1), (3, 2)

Assume background color is (1, 1, 1).

b) Consider the following parameters for an orthographic ray-tracing: [6]

Camera frame: the viewpoint e = (2, 0, 0) and the viewing direction is upward.

Image plane: l = −8, r = 8, t = 8, b = −8

Raster image resolution: 8 × 8

Sphere: centered at e with a radius of 5 units.

Determine the ray-sphere intersection points (if any) for a ray (with length = 12 units) originating at pixel (3, 5) on the raster screen.

c) A curve is characterized by the following rules: [6]

Variables: F, X
Constants: [,], +, −
Axiom: X
Production rules:
X → F − [[X] + X] + F[+FX] − X,
F → FF
Angle: 45°

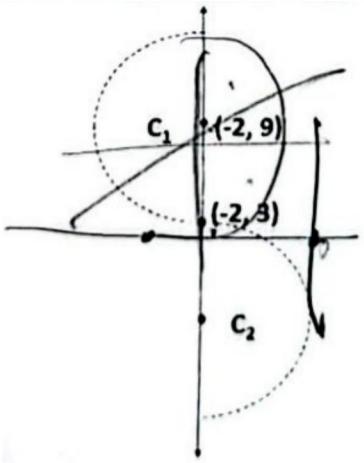
Here, F means "draw forward", X means "do nothing", + means "turn left by the angle", − means "turn right by the angle", [means "push current position and angle" and] means "pop position and angle". Apply the concept of L-Systems to draw the curve for the second iteration (n = 2).

d) Consider a viewport of size 512 × 256 with its origin at the center. Given a point P(−3, 4, −5), transform it to the viewport from an orthographic view volume of dimensions 10 × 12 × 12, centered at the origin. [6]

e) Consider a point P(1, 3, 4) in 3D space. Apply a quaternion-based rotation to the point by 45° around the axis defined by the line y = 3x. [6]

f) Consider a rectangle with vertices A(1, 1), B(6, 1), C(6, 5) and D(1, 5). Apply appropriate transformation to the rectangle to obtain a parallelogram so that point C and D move 12 units to its right from the original position point A and B remain fixed. Find the final vertices after the transformation. [6]

g) In the following figure, there are 2 half-circles C_1 and C_2 where C_2 is the reflection of C_1. Here C_1 has a center at (−2, 9) and both share the same point (−2, 3) on their edge. Using Bresenham's circle drawing algorithm, find the points in the circumference for C_2. [6]



Date of Examination: 17/04/2025

Ahsanullah University of Science and Technology
Department: Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering
Semester Final Examination: Spring 2024
Year: 4th Semester: 2nd
Course Number: CSE4203
Course Title: Computer Graphics

Time: 3 (Three) hours **Full Marks: 70**

Instruction: There are three sets of questions. Answer any five questions from each set. Marks allotted are indicated in the margin.

Question 1: Answer any 5 questions. **[3×5=15]**

a) Explain how a transmissive device works with an example. [3]

b) Describe how the appearance of the object would change for different values of Phong exponent in the Phong shading model. [3]

c) Describe the role of the orthographic transformation in the rendering pipeline. [3]

d) State the differences between the Bezier Curve and the B-spline curve. [3]

e) Explain the programmable stages of the graphics pipeline. [3]

f) Briefly explain the properties of affine transformation with examples. [3]

g) Describe the backface culling technique and how it optimizes the rendering pipeline. [3]

Question 2: Answer any 5 questions. **[5×5=25]**

a) Show that the result of two successive translations $T(tx_1, ty_1)$ and $T(tx_2, ty_2)$ is equivalent to a single translation $T(tx_1 + tx_2, ty_1 + ty_2)$. [5]

b) Consider an open uniform quadratic B-spline curve S defined by 7 control points $P_0(-2, 1)$, $P_1(-1, 2)$, $P_2(0, 3)$, $P_3(1, 4)$, $P_4(2, 3)$, $P_5(3, 2)$ and $P_6(4, 1)$. Determine the point on the curve segment for $S_3(1/5)$. [5]

c) Derive the OpenGL perspective projection matrix M_OpenGL from the general perspective matrix and orthographic transformation matrix. [5]

d) Consider a quadrilateral ABCD with vertices A(1, 3), B(5, 3), C(8, 5) and D(4, 7). Using the concept of barycentric coordinate, determine if a point P(6, 3.5) is inside the quadrilateral or not. [5]

e) Show that frame-to-canonical transformation can be expressed as a rotation followed by a translation. [5]

f) Consider 2 images IMG1 and IMG2 with the pixel values shown below. Each pixel in the image can hold up to 8 bits of data. Determine which image has the better compression, if the images are compressed by applying run-length compression. Show necessary calculations to compare the compression. [5]

255	255	128	128	128
128	128	128	255	255
255	255	255	0	0
255	255	0	0	0
255	255	128	128	128

IMG1

255	128	128	128	255
128	128	255	255	255
255	128	128	128	255
255	255	0	255	0
255	255	0	255	0

IMG2

g) Find the dimension of a fractal generated by the following steps: [5]

Step 1: Start with a square.

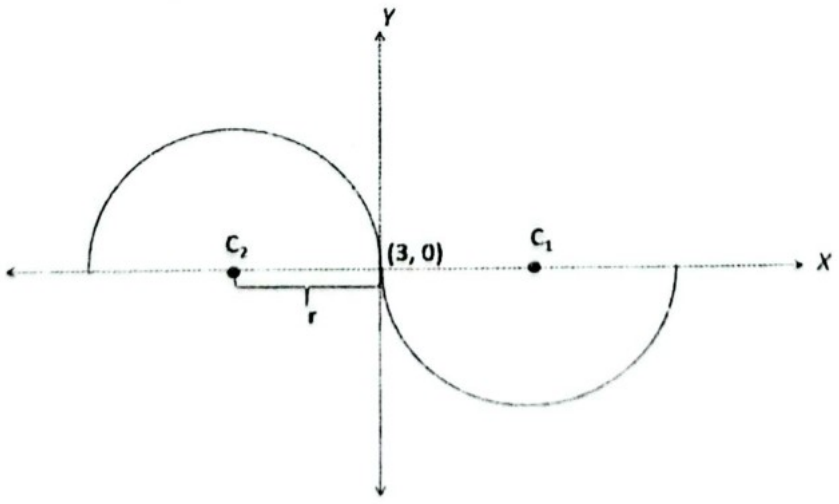
Step 2: Divide it into 9 equal squares and remove the center square.

Step 3: Repeat for each remaining square at the next iteration.

Question 3: Answer any 5 questions. **[6×5=30]**

a) Consider a clipping rectangle with a width and height of 12 and 10 units respectively. Its lower left corner is located at (3, 4). Also, consider a line with starting and ending points of (1, 8) and (18, 2) respectively. Perform the line-edge intersecting points for all four edges of the clipping rectangle using Cyrus-Beck algorithm and determine the true clipping points. Show the steps and calculations for your solution (assume any data if necessary). [6]

b) Draw the following pattern using Bresenham's circle drawing algorithm. The pattern is constituted of two half circles with the centers C_1 and C_2 respectively having the same radius r = 5. Both circumferences go through (3, 0). Show the steps to determine the points and plot them. [6]



c) Assume ABCD is a 2D rectangle with vertices A(1, 1), B(10, 1), C(10, 7) and D(1, 7). Transform ABCD to obtain A′B′C′D′ such that A′D′ and B′C′ both create 45 degrees with the X-axis. Determine the composite transformations and find the final vertices. [6]

d) A curve is characterized by the following rules: [6]

Variables: F

Constants: [,], +, −

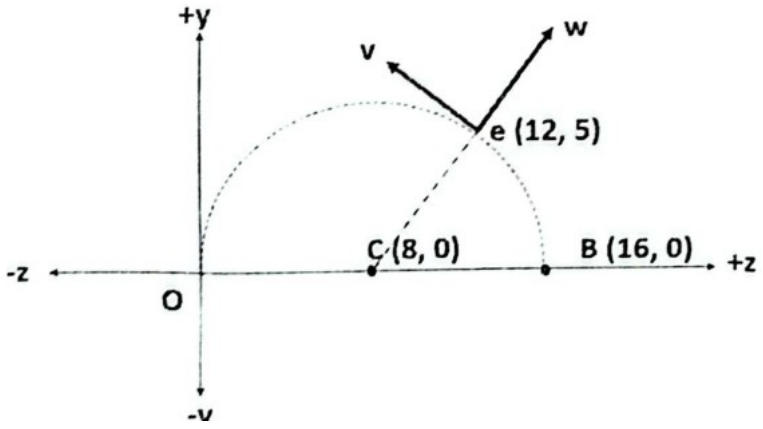
Axiom: F

Rules: F → F[+F][−F]F

Angle: 45°

Here F means “draw forward”, + means “turn left by angle”, − means “turn right by angle”, [means “push current position and angle” and] means “pop position and angle”. Apply the concept of L-systems to draw the curve for the third iteration (n = 3).

e) Origin O and basis {z, y} construct a 2D canonical coordinate system (see figure) where −z is the viewing direction and y is the up vector. A drone with eye position e and basis {w, v} is designed to move along a half-circle path with center C while facing towards point C. Determine the position of points B and C in the drone’s coordinate system. [6]



f) Consider a 3D line AB with vertices A(−1, −1, −1) and B(1, 1, 1) defined within a canonical view volume of dimension 4 × 4 × 4 (centered at the origin). Construct the viewport matrix to transform the line to a viewport of size 512 × 256 and determine the corresponding pixel coordinates. [6]

g) Consider a rectangle with vertices A(10, 6), B(16, 6), C(16, 10) and D(10, 10). Reflect the rectangle along the line 2y − 3x − 4 = 0 using 2D transformation and find the final vertices. [6]